

4(a) The increase in the internal energy of a system is the sum of the external **B1**
work done on the system and the heat supplied to the system.

4(b)(i) Using $T = \frac{pV}{nR}$ **M1**

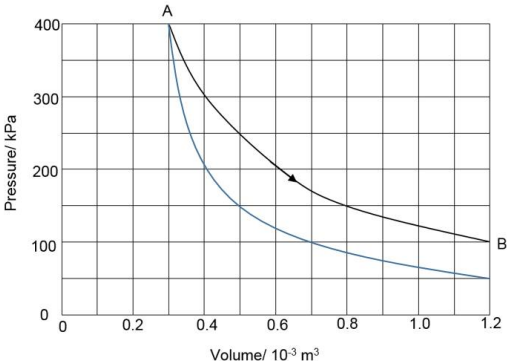
Taking any point:

$$\text{Point A: } T_A = \frac{(400 \times 10^3)(0.3 \times 10^{-3})}{(0.050)(8.31)} = 289 \text{ K}$$

$$\text{Point B: } T_B = \frac{(100 \times 10^3)(1.2 \times 10^{-3})}{(0.050)(8.31)} = 289 \text{ K}$$

Therefore, temperature is 289 K. **A1**

4(b)(ii) Kinetic energy $= \frac{3}{2} nRT = \left(\frac{3}{2}\right) (0.050)(8.31)(289) = 180 \text{ J}$ **M1**
- **A0**

Question		Marks
4(b) (iii)	Work done by gas = area under the p - V graph = (25 squares+9 missing squares) $(50 \times 10^3) (0.10 \times 10^{-3})$ = 170 J (allow 165 to 175J)	M1 A1
4(b) (iv)	temperature unchanged, no increase in the internal energy. $\Delta U = 0$ J	
	When the gas expands from A to B, work done on the gas is $w = -170$ J.	M1
	Using 1 st Law of Thermodynamics: $\Delta U = w + q$ $0 = -170 + q$ Heat supplied, $q = +170$ J	A1
	q is positive $\rightarrow$ heat is supplied to the gas during the expansion of the gas to ensure that there is no increase in the internal energy of the gas. Therefore, the student claim is false.	
4(c)	 Downwards sloping curve below AB Ends at (1.2, 50)	B1 B1

5(a) The induced e.m.f. is directly proportional to the rate of change of magnetic flux. B1